

The association between sterilizing activity and drug distribution into tuberculosis lesions

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Evidence abstract

Finding new treatment-shortening antibiotics to improve cure rates and curb the alarming emergence of drug resistance is the major objective of tuberculosis (TB) drug development.

Using a matrix-assisted laser desorption/ionization (MALDI) mass spectrometry imaging suite in a biosafety containment facility, we show that the key sterilizing drugs rifampicin and pyrazinamide efficiently penetrate the sites of TB infection in lung lesions.

Rifampicin even accumulates in necrotic caseum, a critical lesion site where persisting tubercle bacilli reside.

In contrast, moxifloxacin, which is active in vitro against a subpopulation of *Mycobacterium tuberculosis* that persists in specific niches under drug pressure and has achieved treatment shortening in mice, does not diffuse well in caseum, concordant with its failure to shorten therapy in recent clinical trials.

We suggest that such differential spatial distribution and kinetics of accumulation in lesions may create temporal and spatial windows of monotherapy in specific niches, allowing the gradual development of multidrug-resistant TB.

We propose an alternative working model to prioritize new antibiotic regimens based on quantitative and spatial distribution of TB drugs in the major lesion types found in human lungs.

The finding that lesion penetration may contribute to treatment outcome has wide implications for TB.

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